# Genomics of Gene Regulation: Concepts and Techniques

**Ana Azambuja**
Senior Research Associate
Pathology Department - Boston Children’s Hospital

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Epigenomic regulation and DNA-protein interaction

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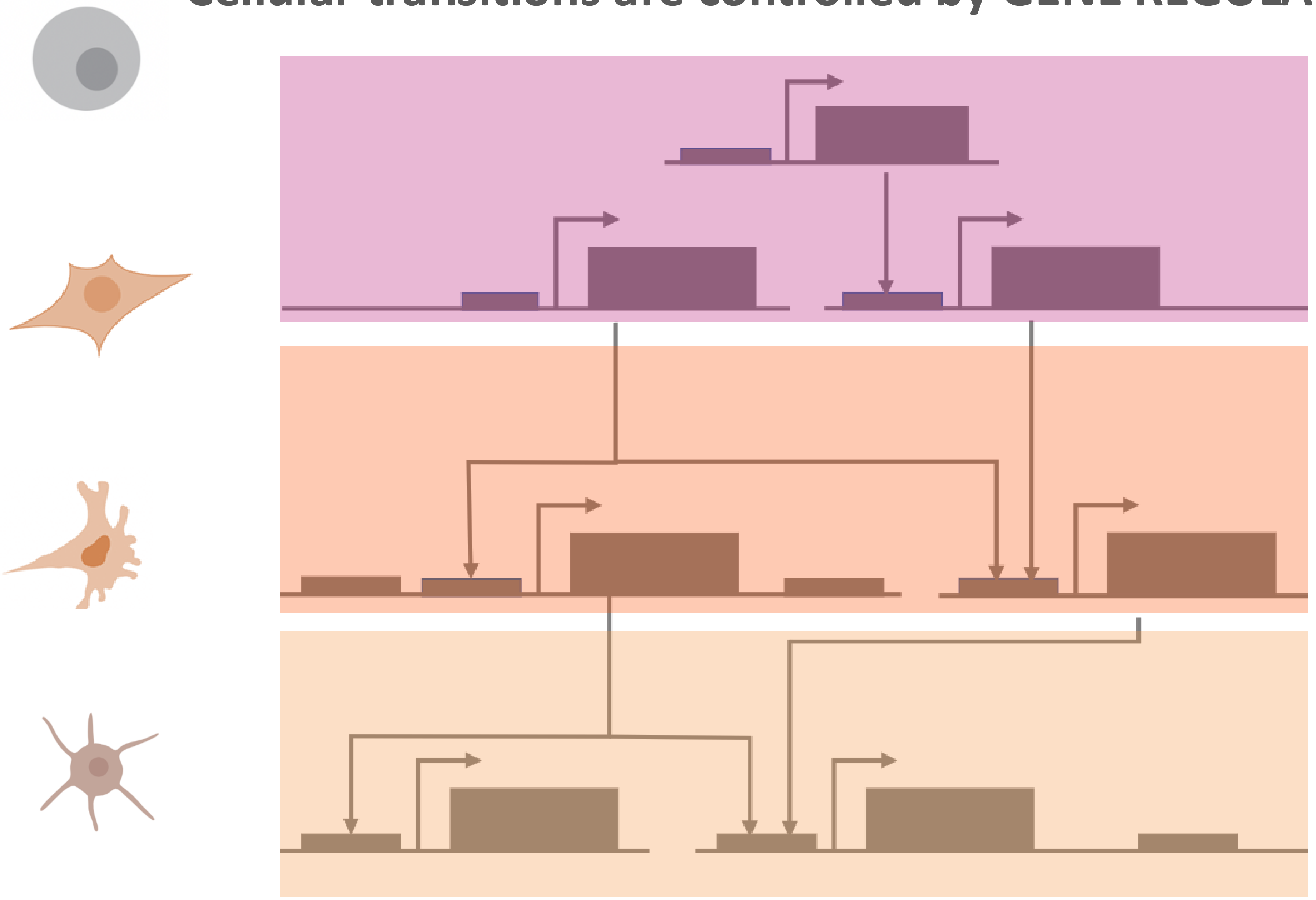
# Cellular transitions are controlled by GENE REGULATORY NETWORKS



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Epigenetic regulation

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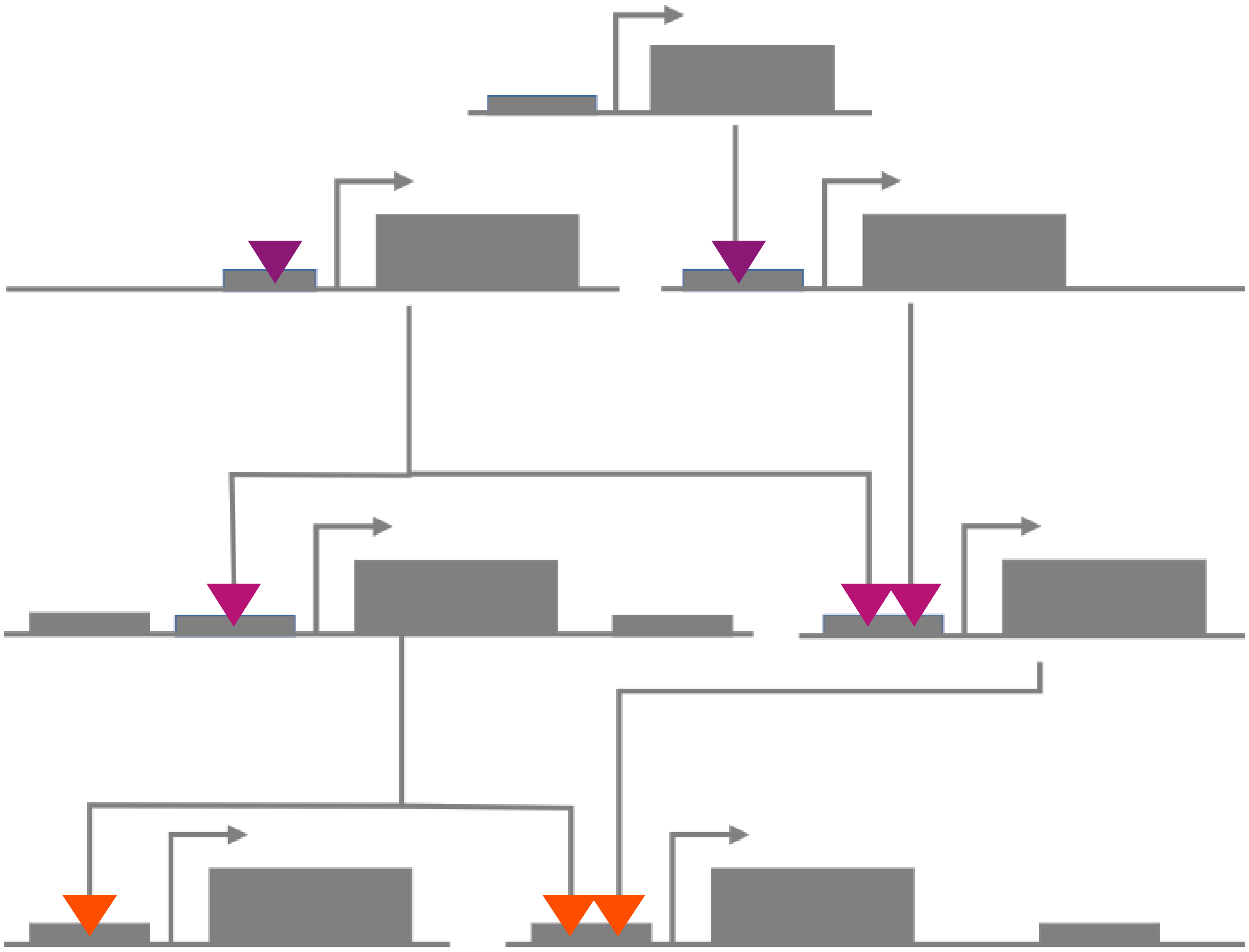
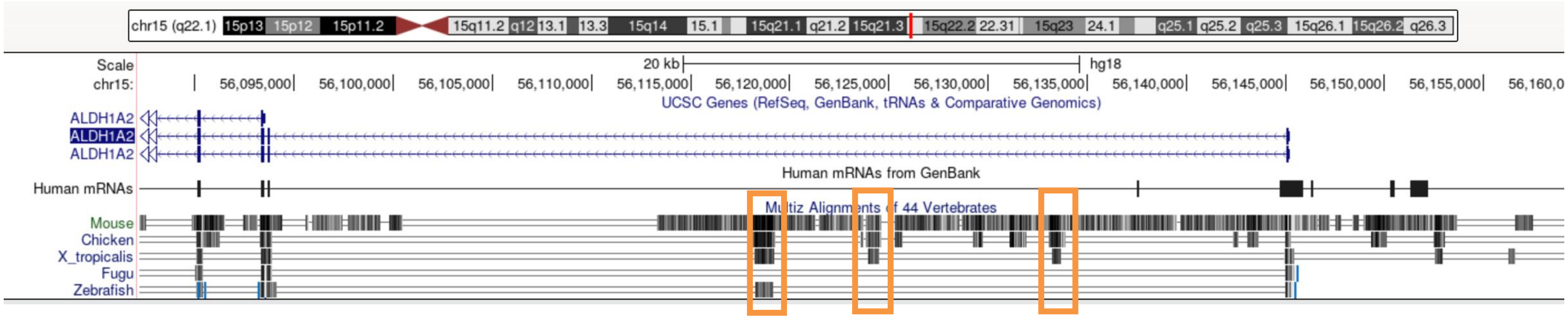
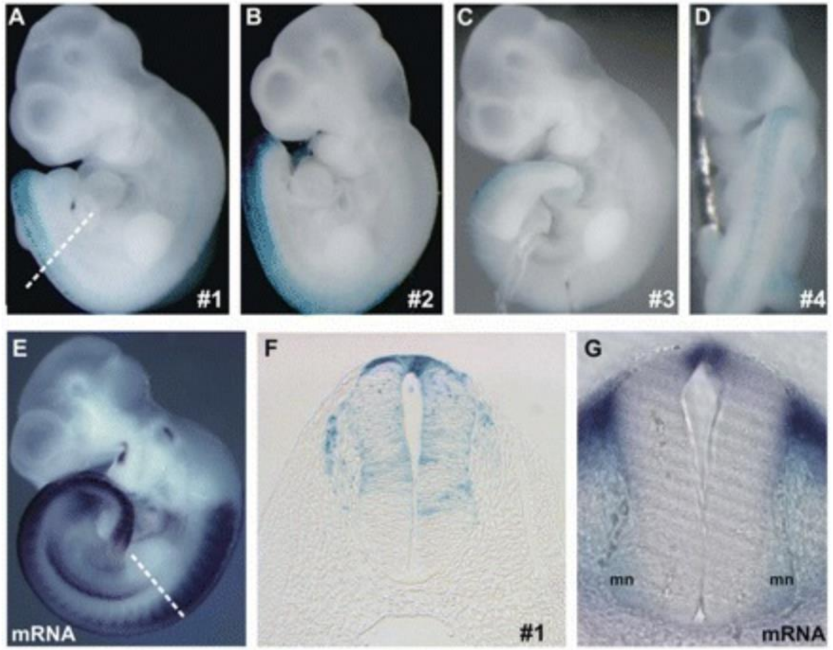
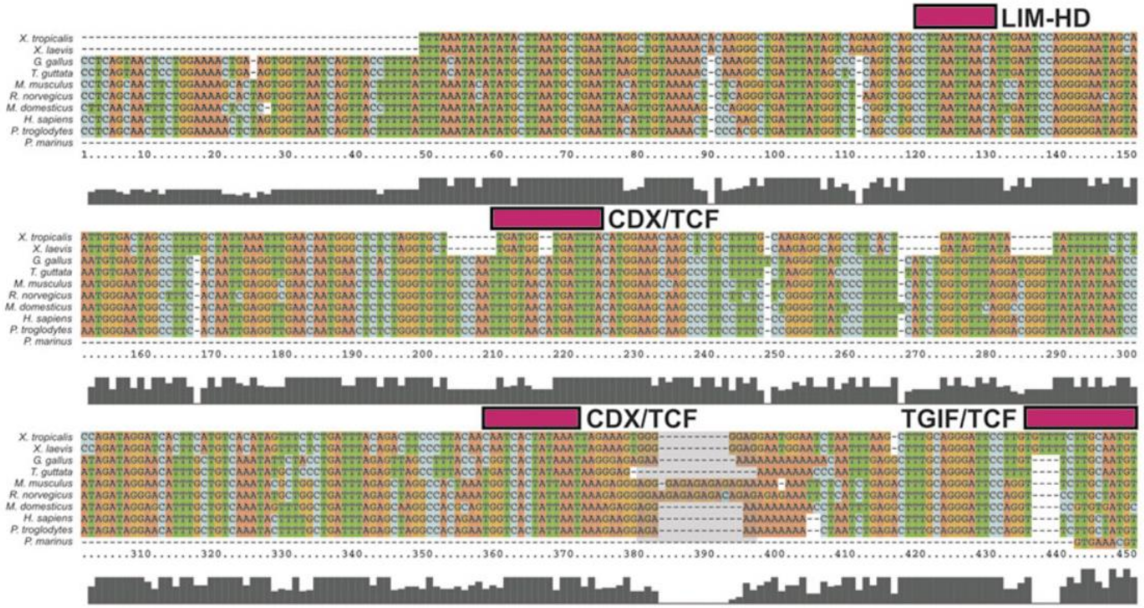


Protein –
DNA interactions

# Cis-regulatory features and how to identify them




Castillo, et al. 2010. Development

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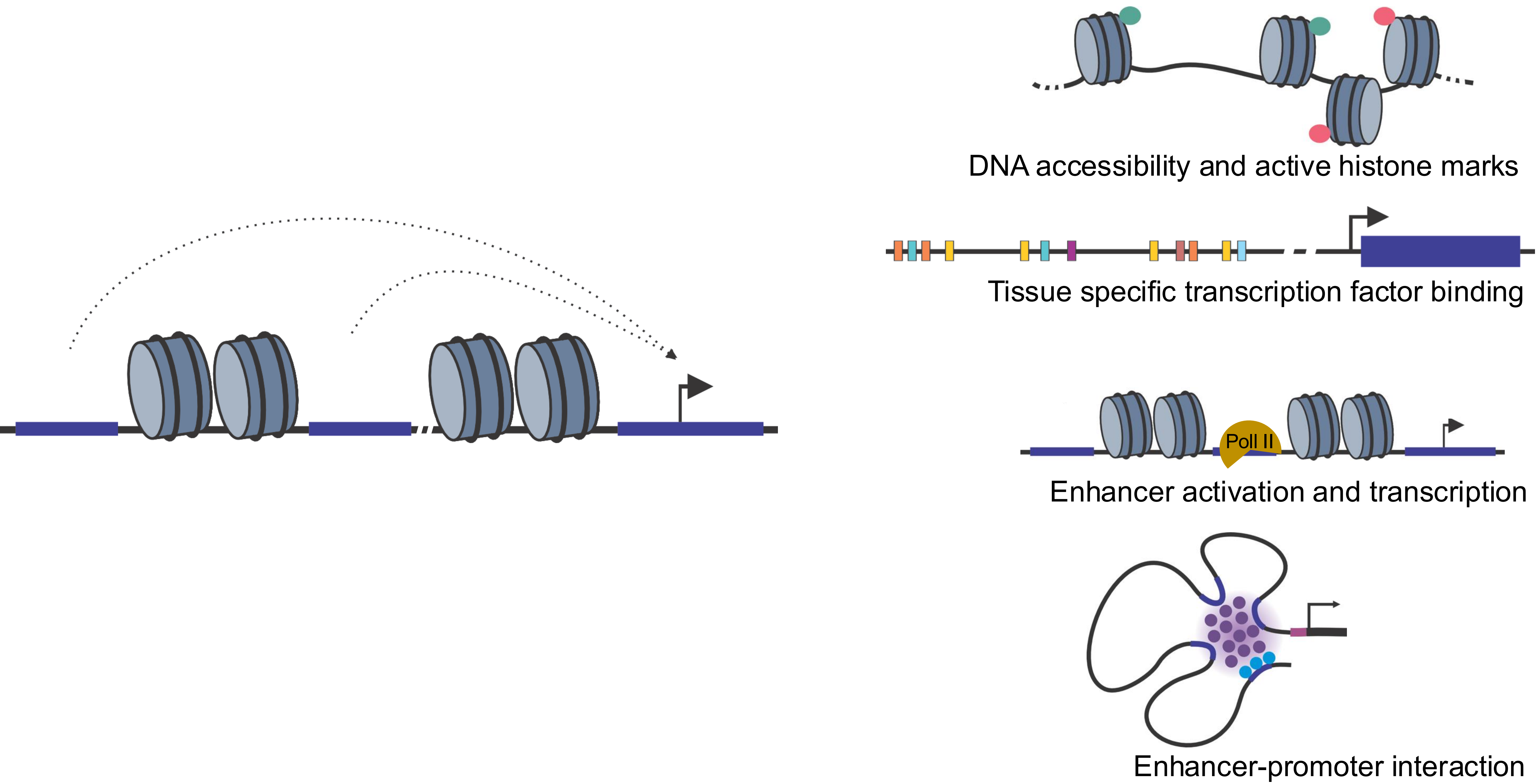
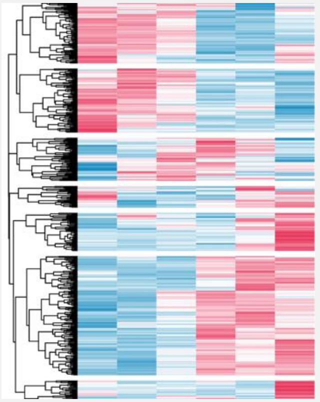


# Topics to be covered - Techniques

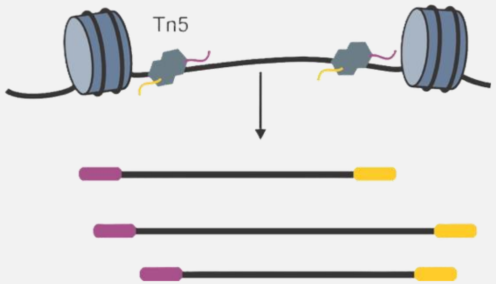
## GENE EXPRESSION



RNAseq

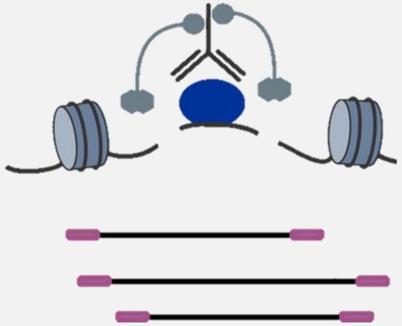
Spatial transcriptomics

## EPIGENOMIC REGULATION




ATACseq

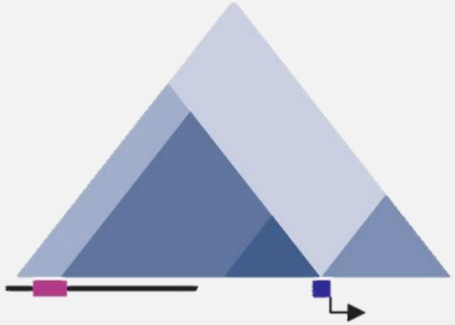
## DNA-PROTEIN BINDING



ChIPseq
CUT&RUN
CUT&TAG

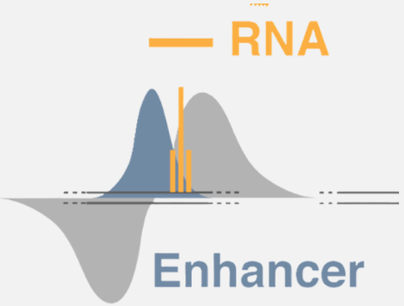
## GENOME ORGANIZATION

3D genome organization - HiC



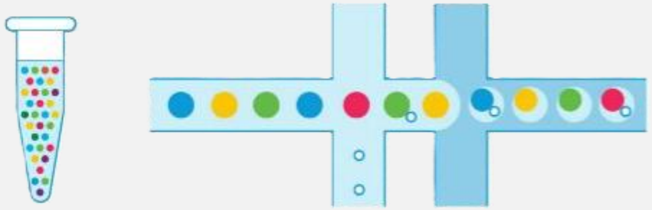
## ENHANCER ACTIVITY

eRNA mapping




## SINGLE-CELL TECHNOLOGIES

Transcriptomics / Multiome



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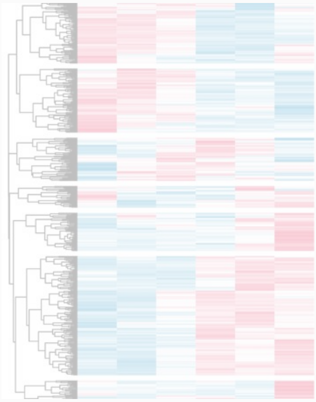

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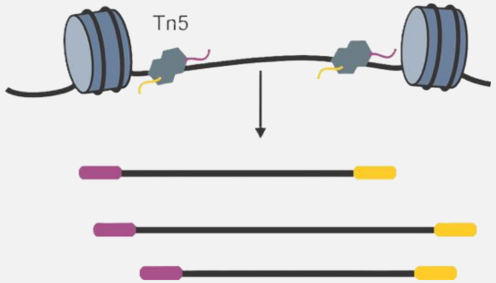


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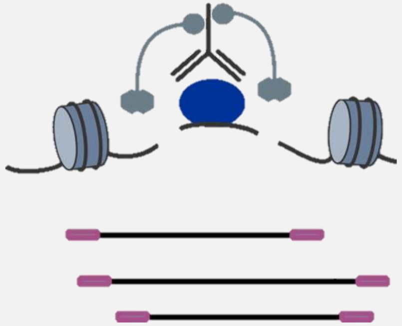

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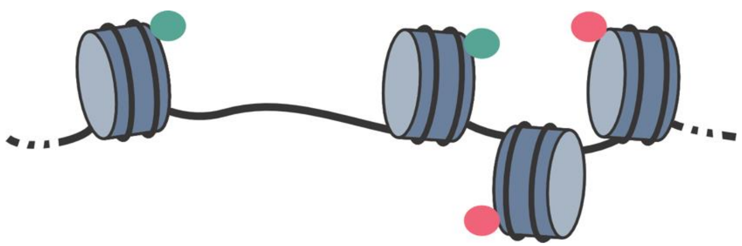

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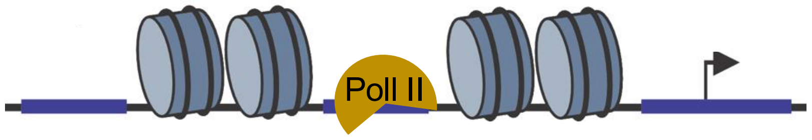


Enhancer activation and transcription

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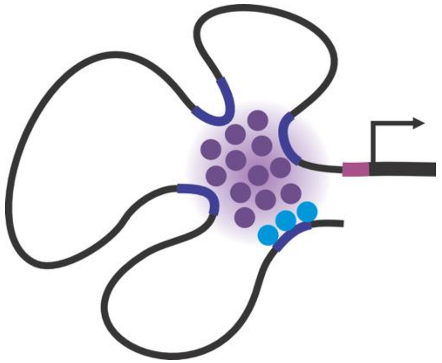
map chromatin contacts genome-wide



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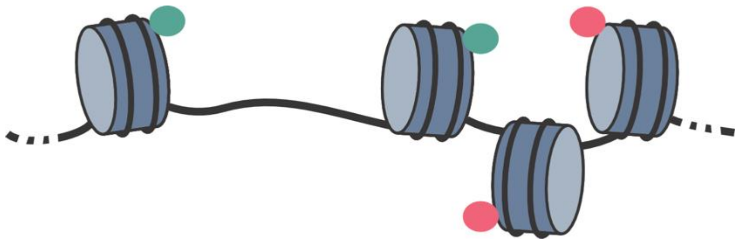

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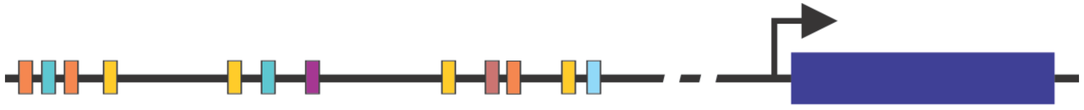

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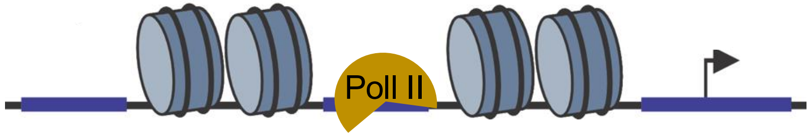


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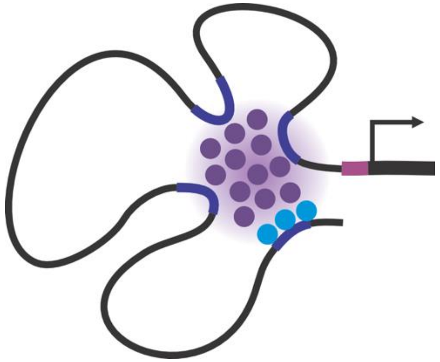

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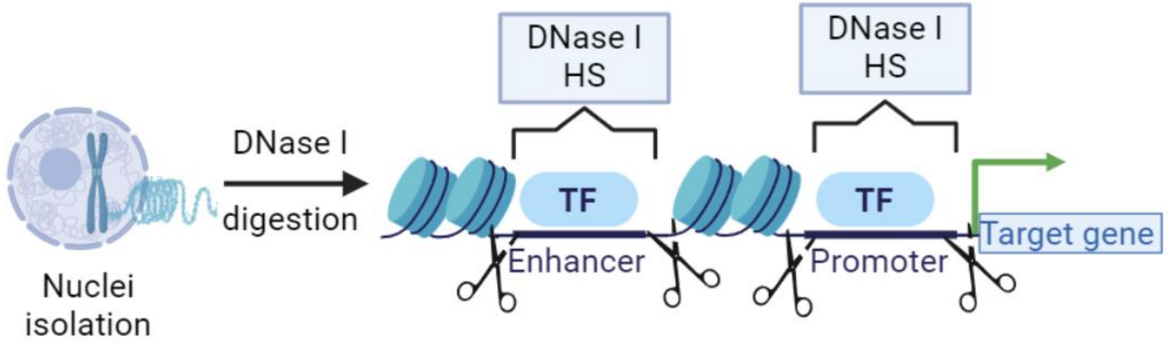
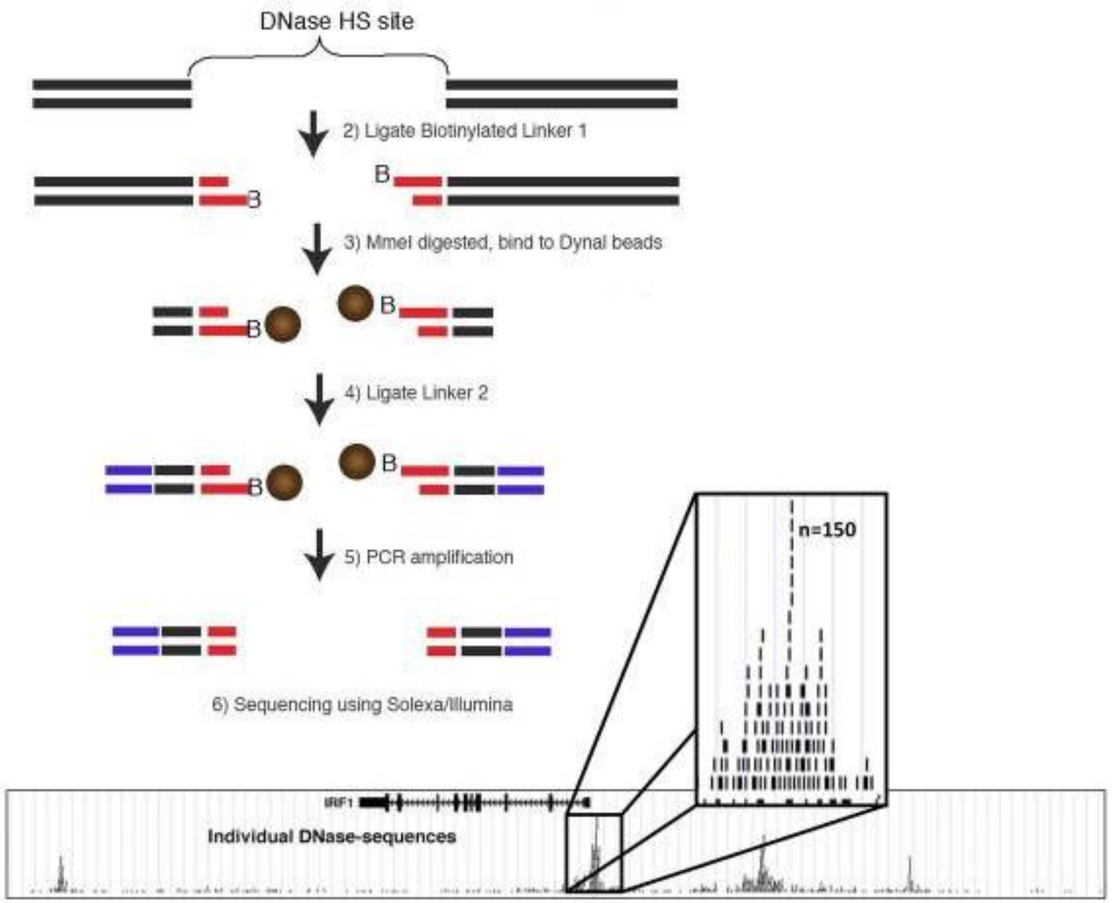
# Mapping DNA accessible regions genome-wide – Dnase-seq




Boyle, et al. 2008. Cell

Minnoye, et al. 2021. Nat Reviews

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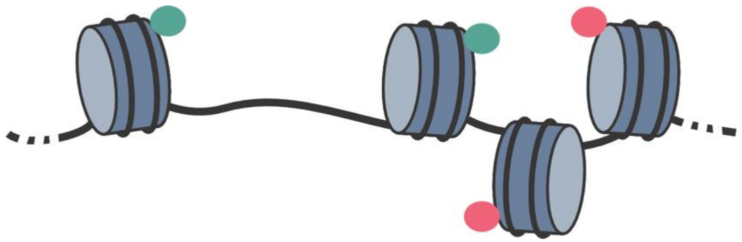

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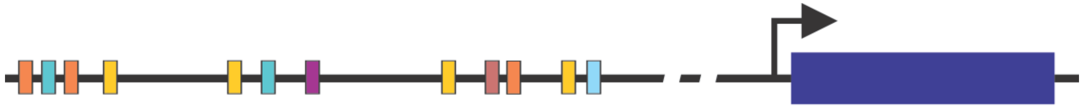

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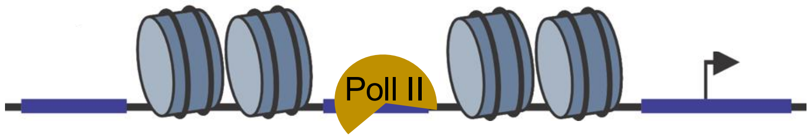


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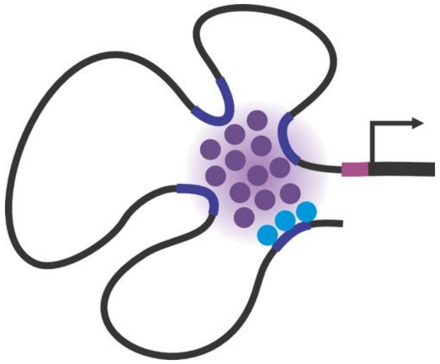

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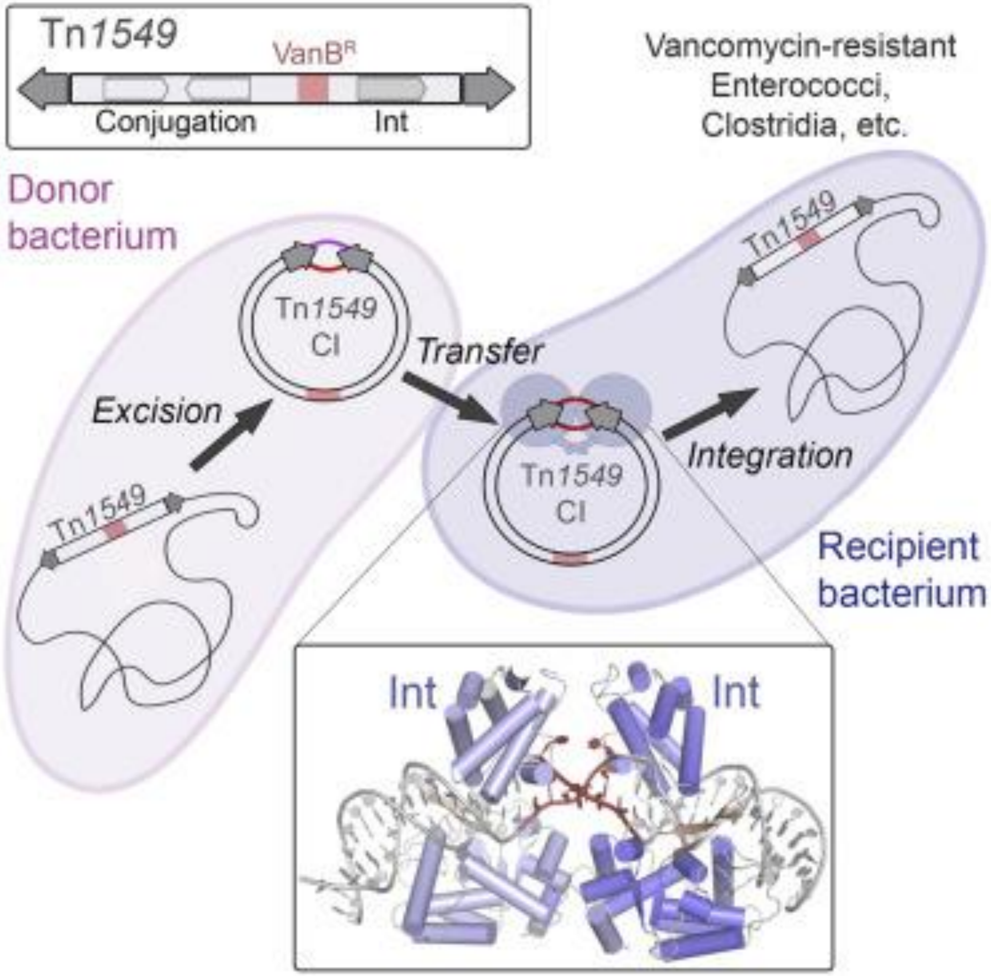


Rubio-Cosials, et al. 2018. Cell

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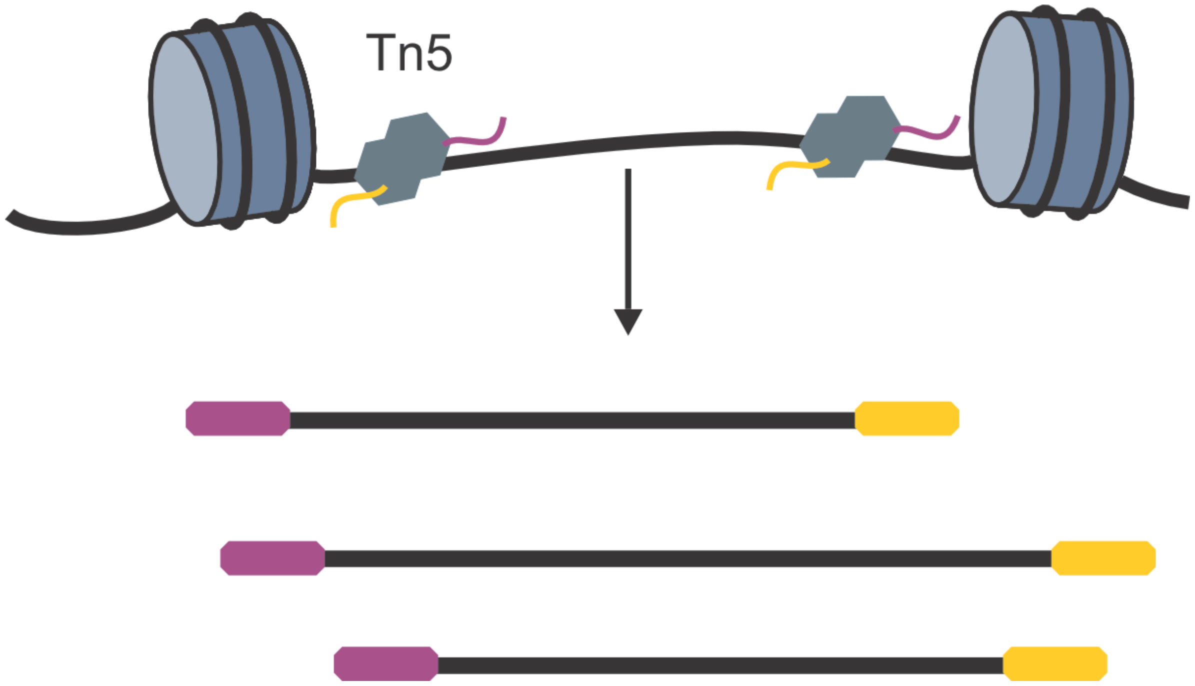
## Assay for Transposase-Accessible Chromatin using sequencing




Buenrostro, et al. 2013. Nat Methods

For review: Minnoye, et al. 2021. Nat Reviews

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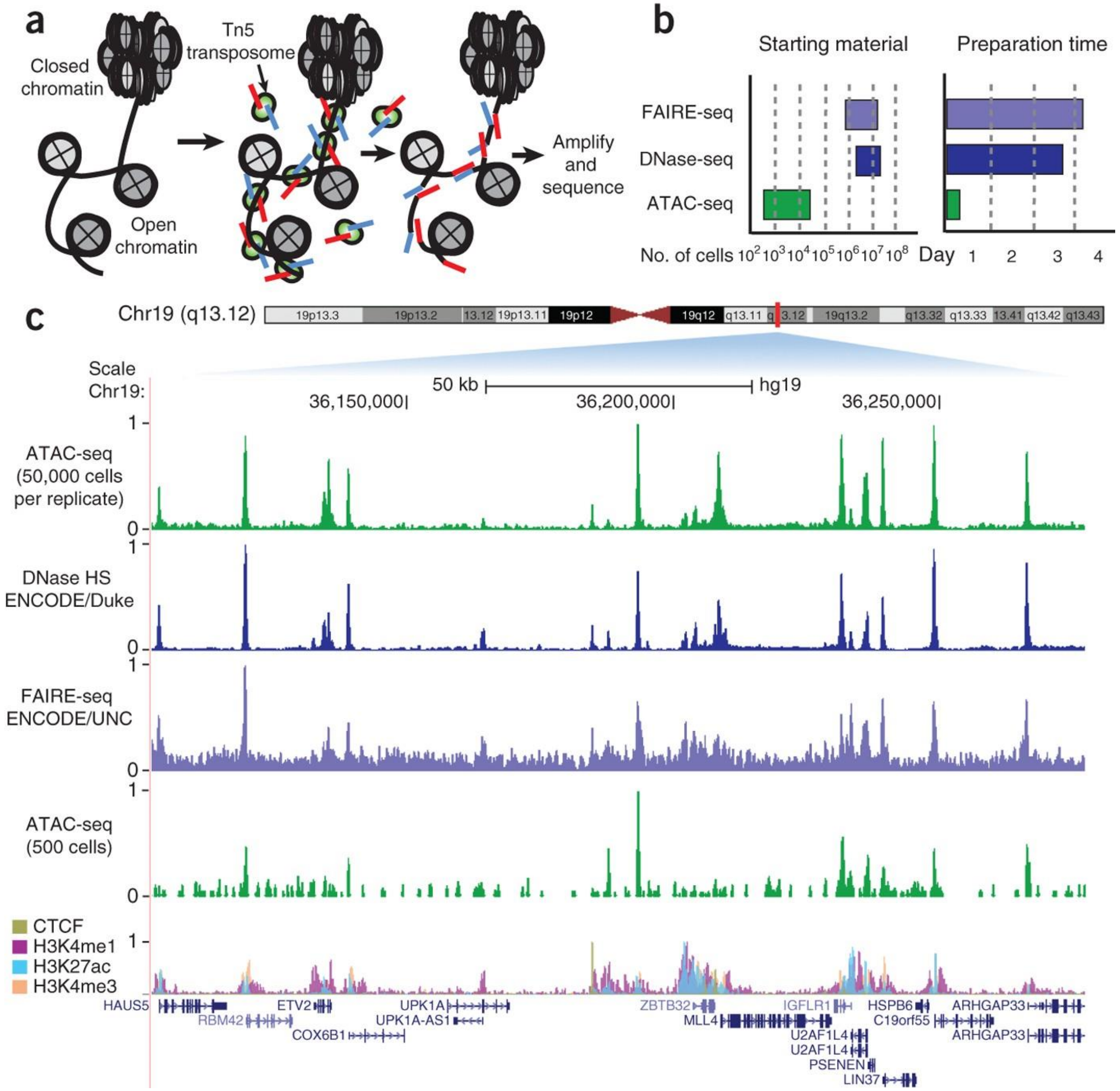


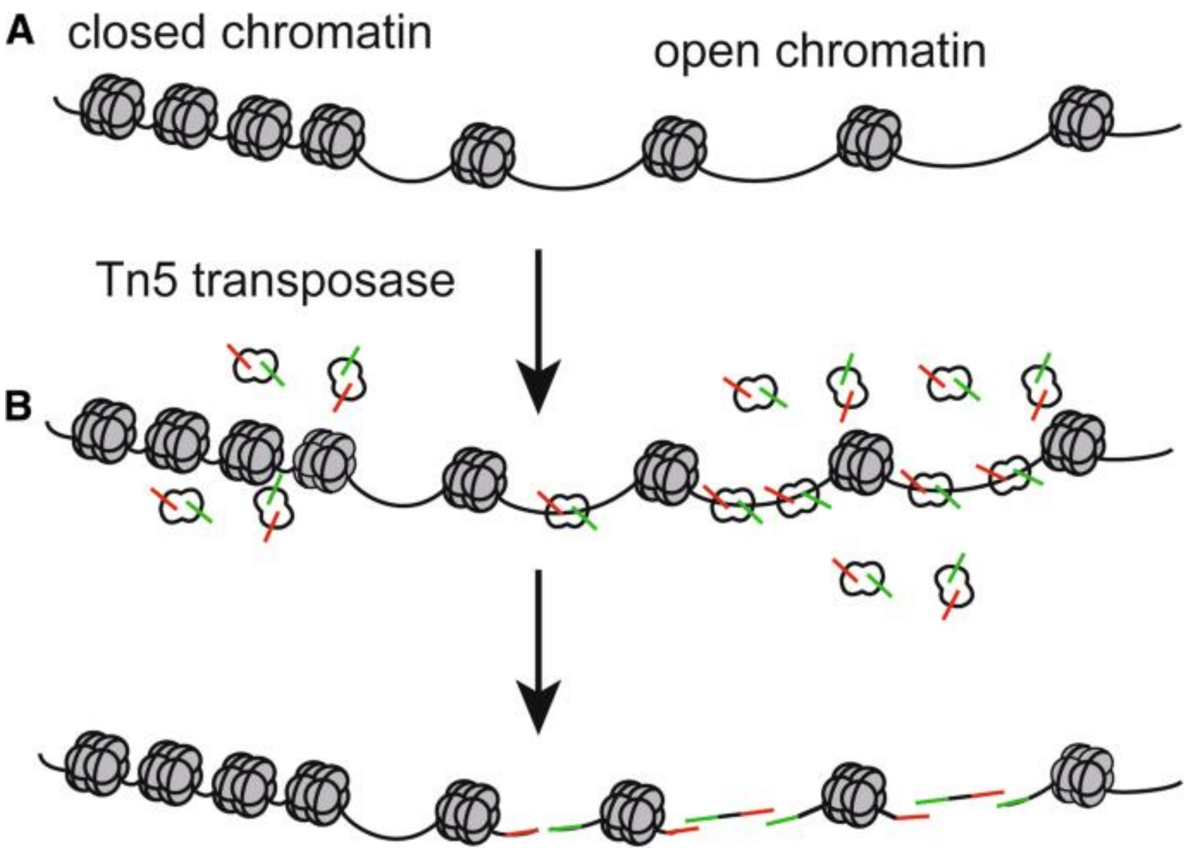


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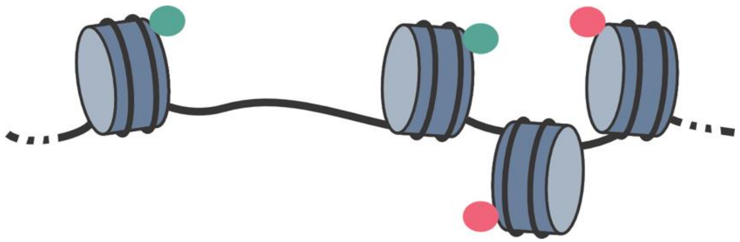

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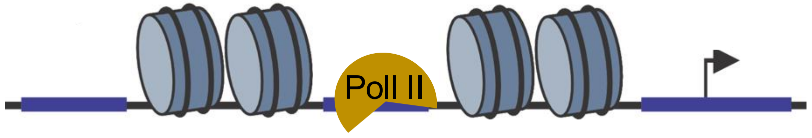


Enhancer activation and transcription

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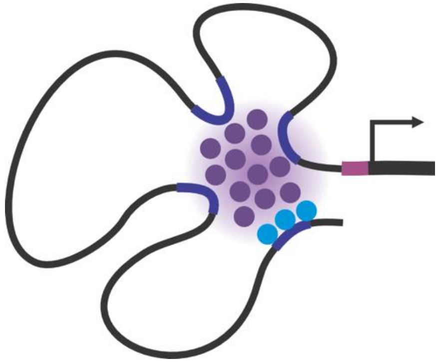

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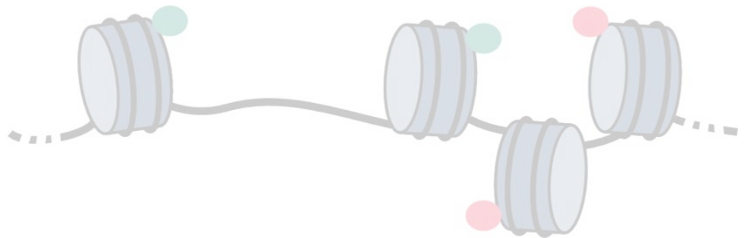

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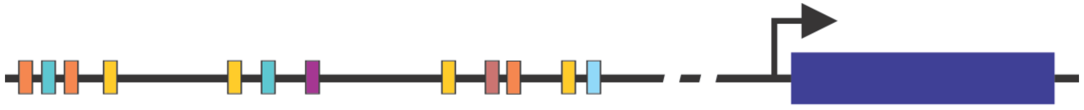

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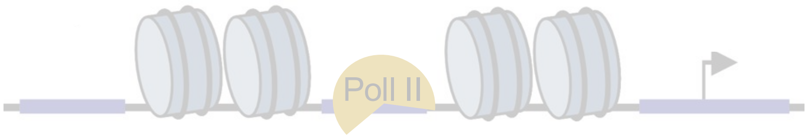


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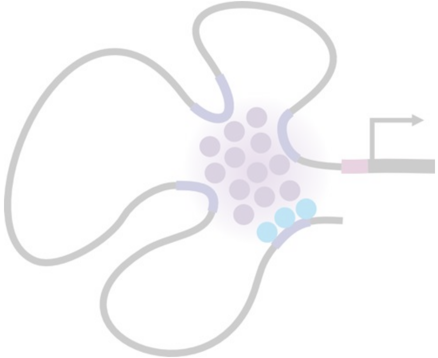
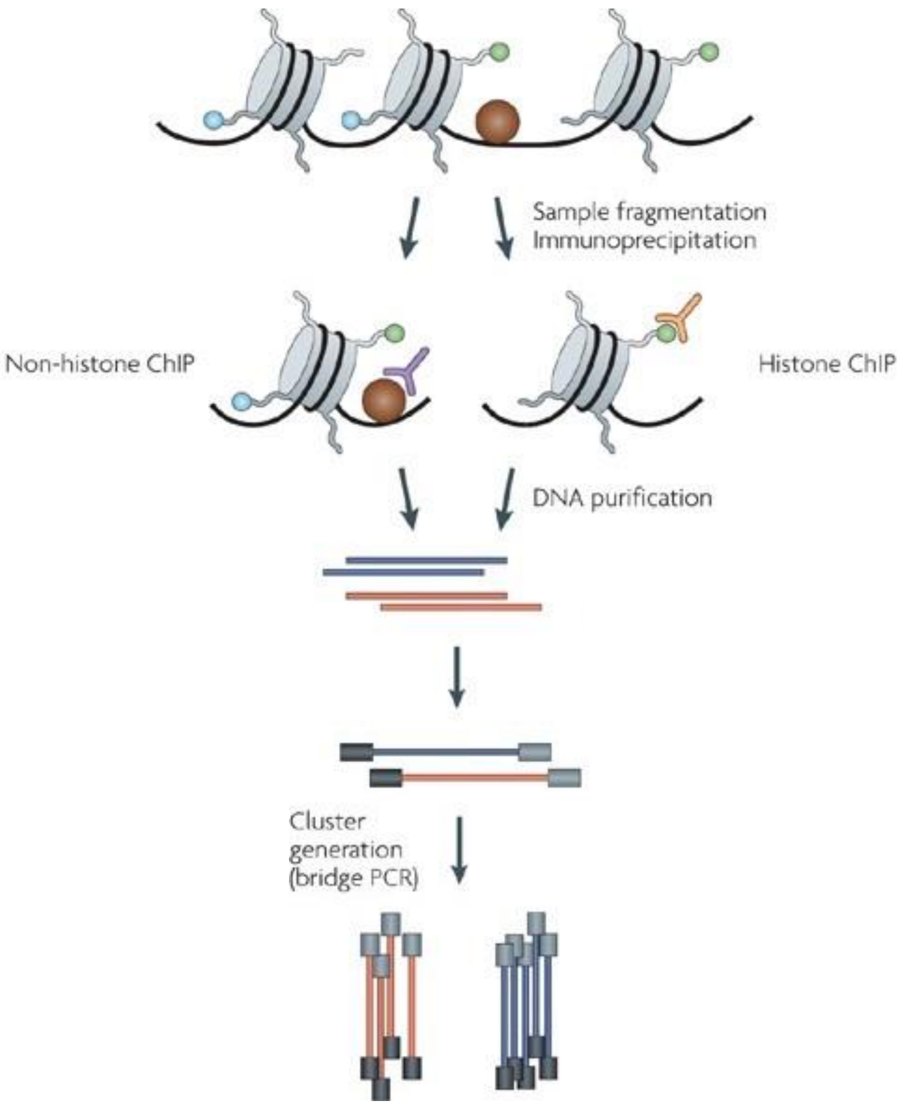
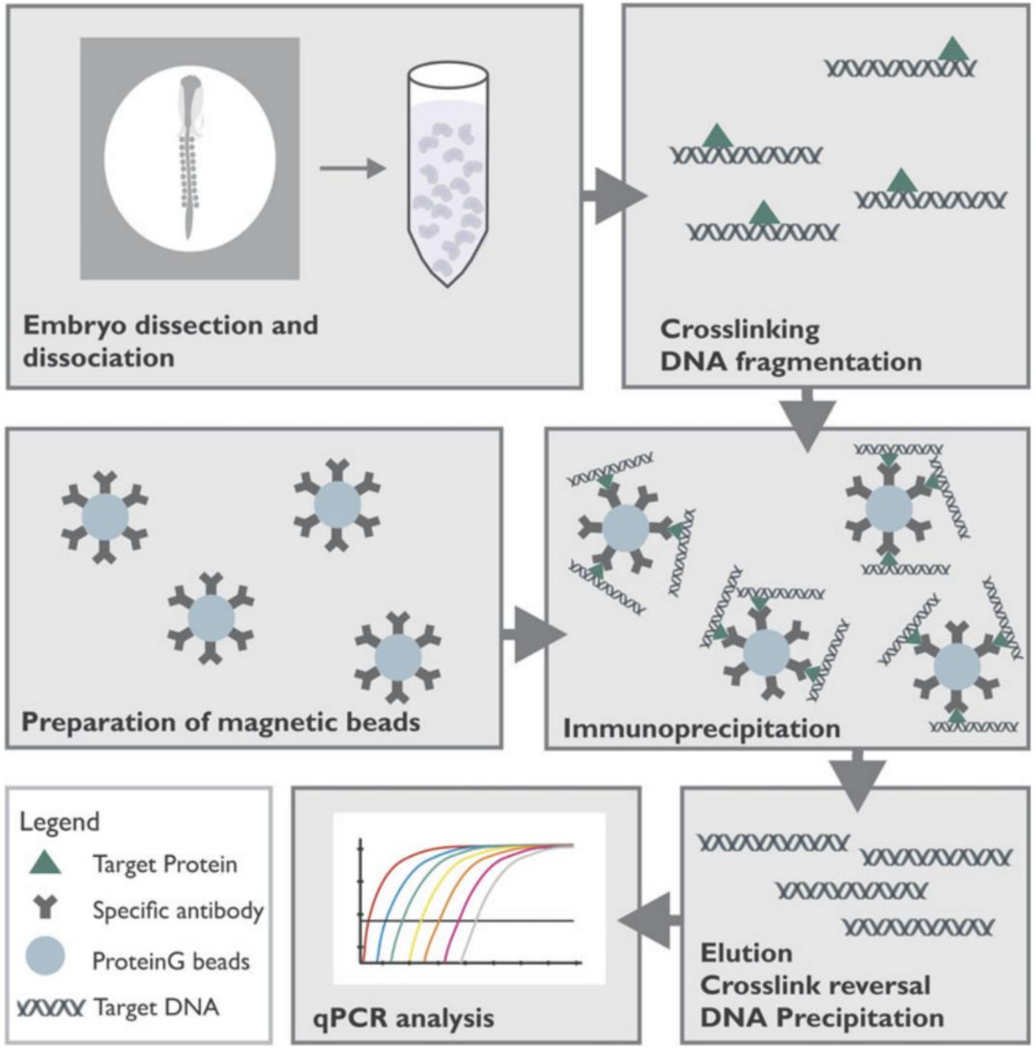


Enhancer-promoter interaction

# Mapping protein-DNA interactions – ChIP-seq
## Chromatin Immune Precipitation - sequencing




Johnson, et al. 2007. Science

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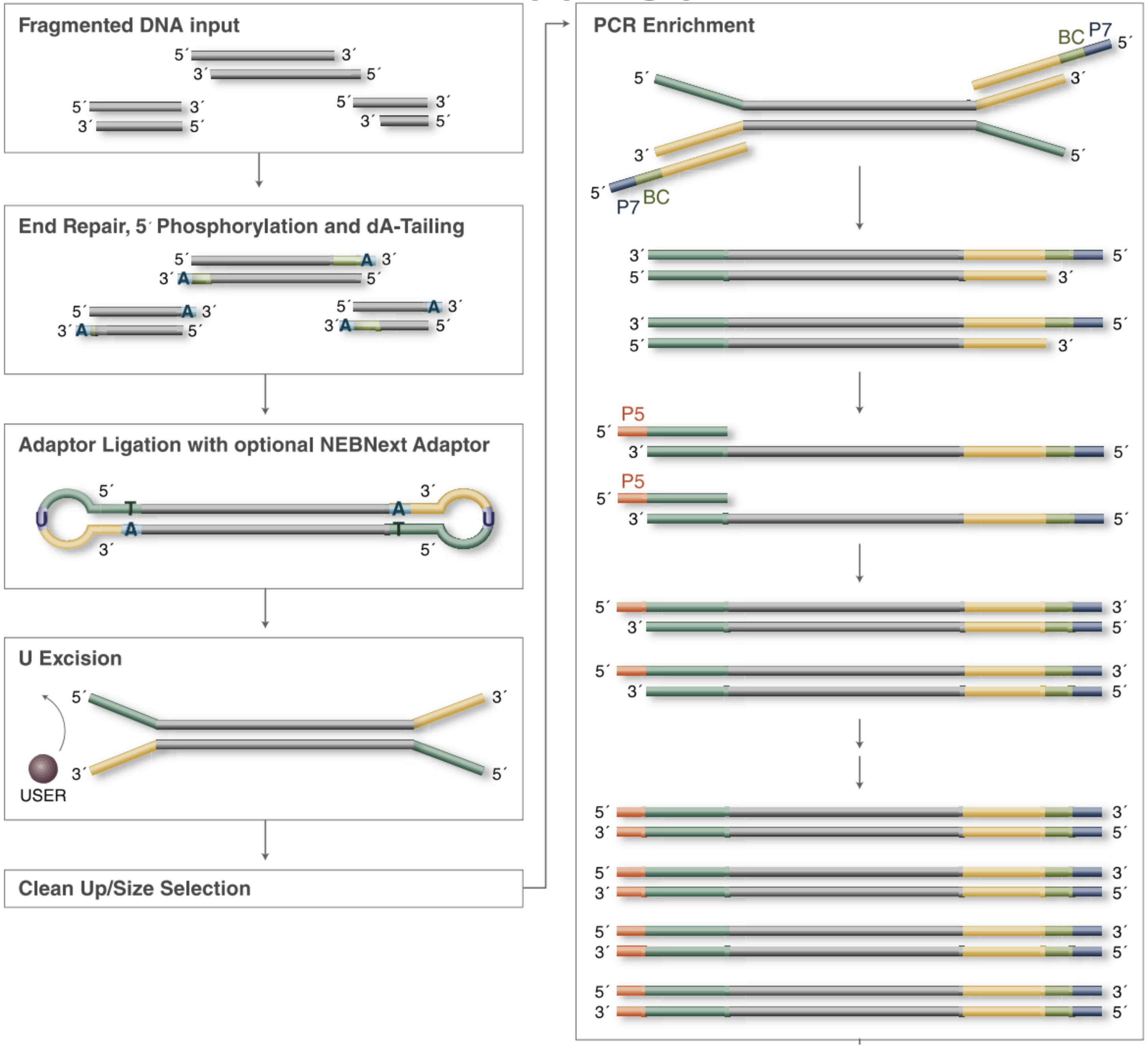
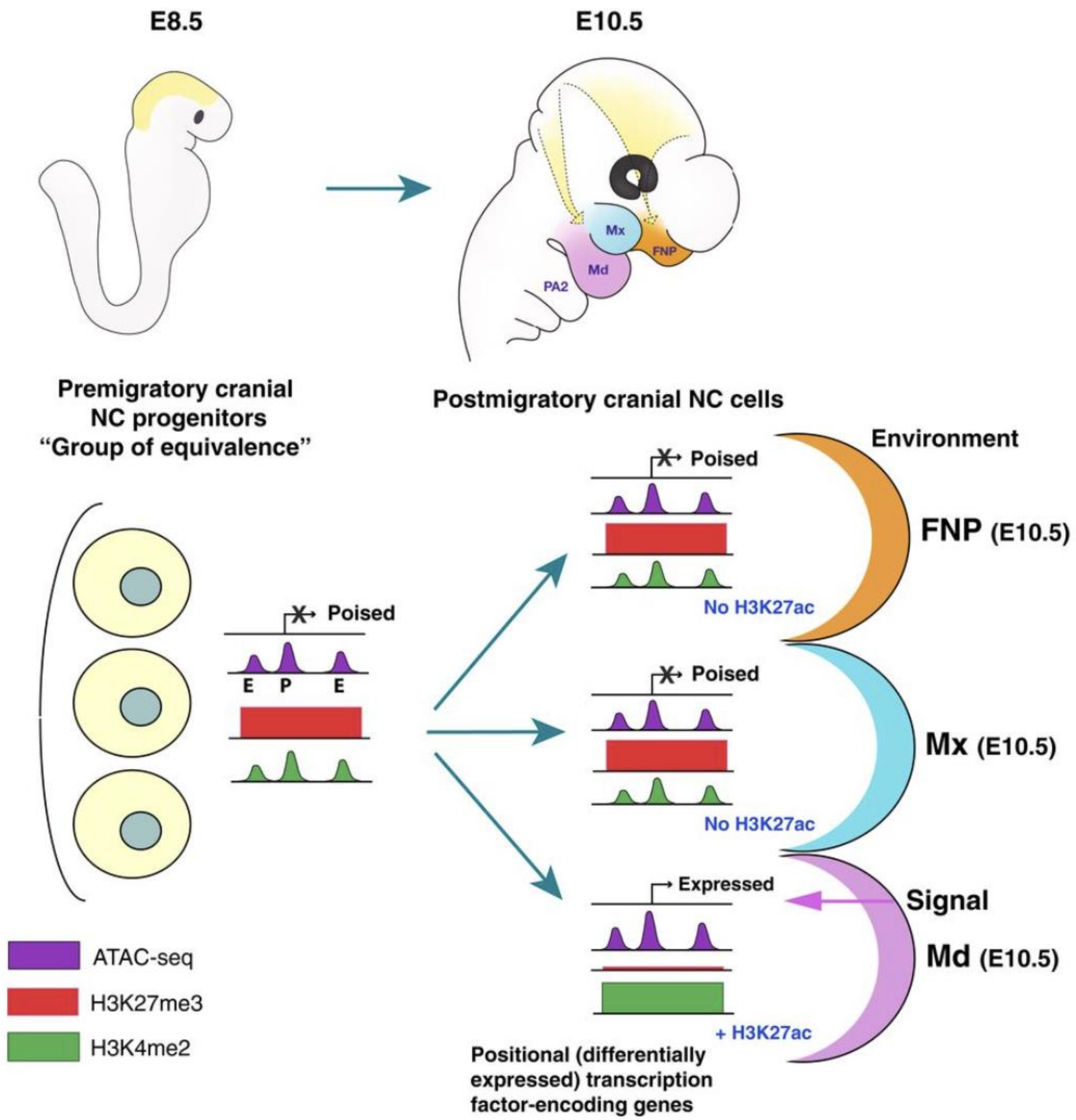


# The power of combining technologies




Minoux. 2017. Science

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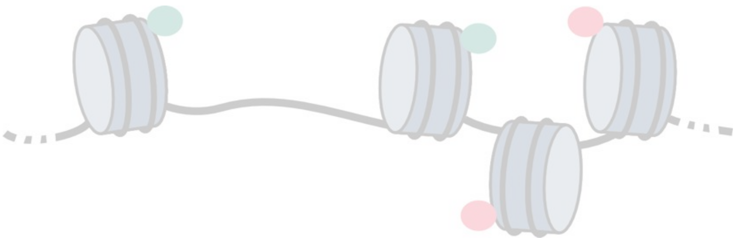

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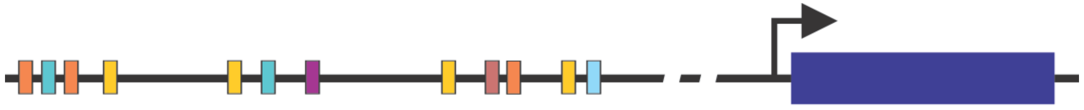

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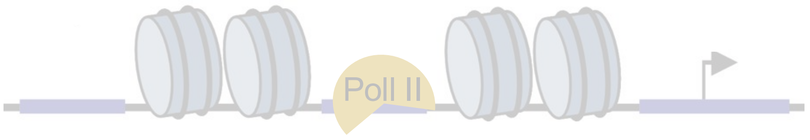


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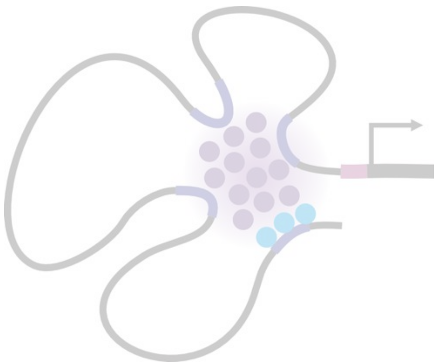
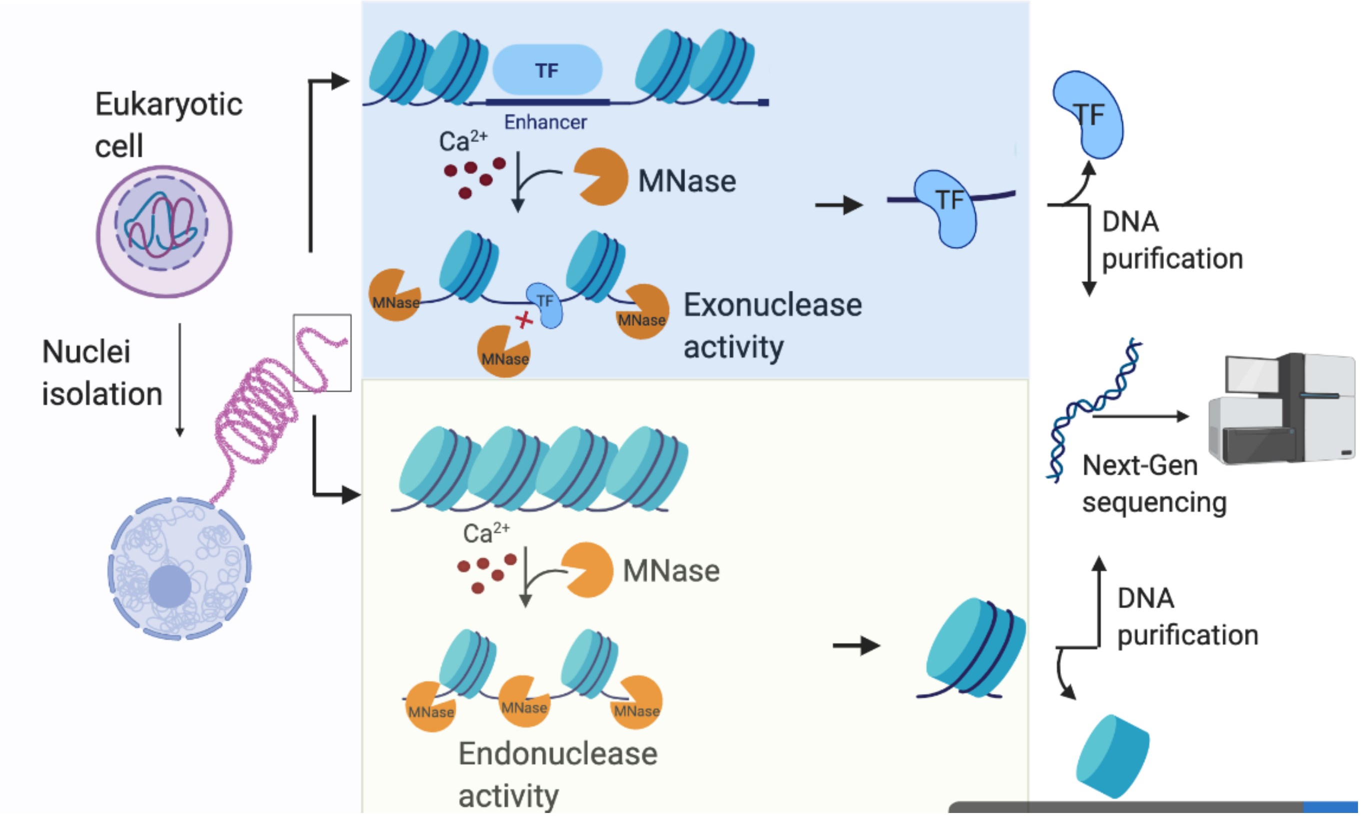


Enhancer-promoter interaction

# Mapping protein-DNA interactions – CUT&RUN

## Cleavage Under Targets & Release Using Nuclease

Mnase - non-specific endo-exonuclease micrococcal nuclease




**Ch**romatin **I**mmuno**c**leavage /
**ch**romatin **e**ndogenous **c**leavage

**ChIC ChEC**

Schmid. 2004. Mol Cell

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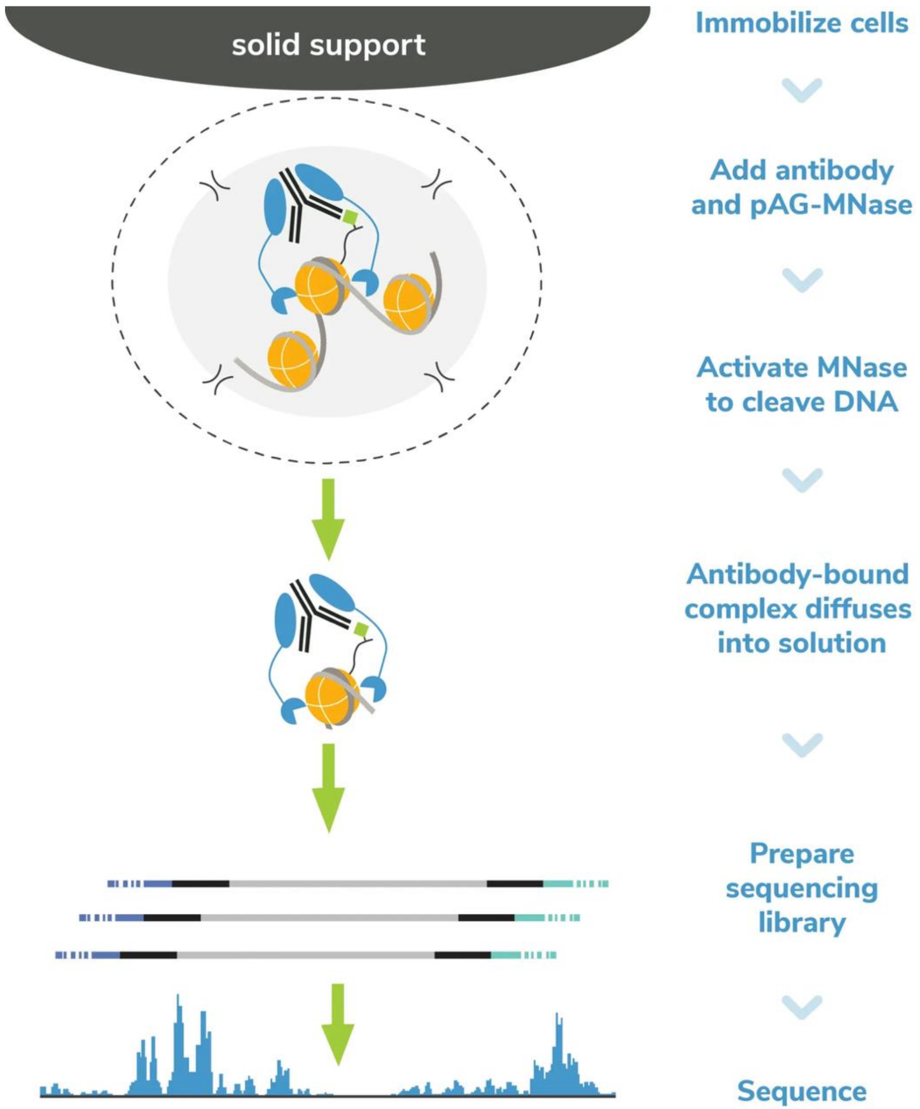
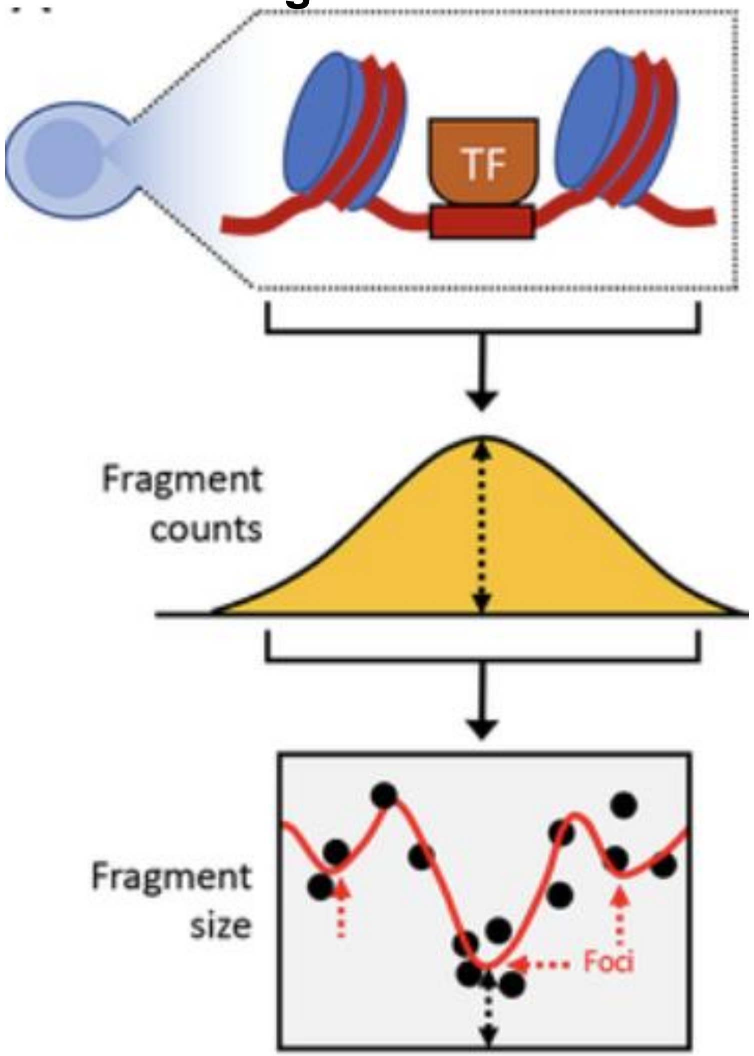



No crosslinking
No sonication
Low input

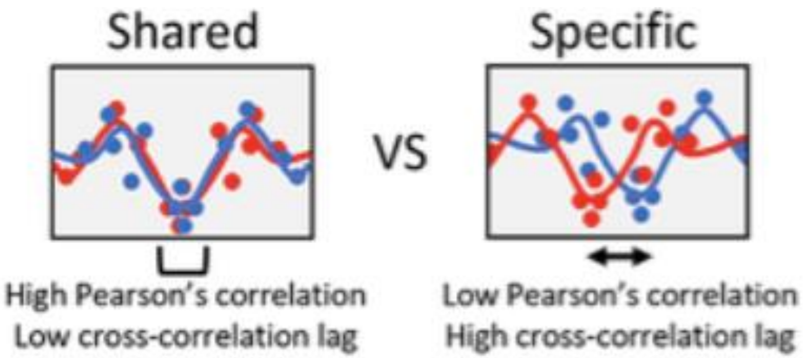
Skene and Henikoff. 2017. eLife

# Mapping protein-DNA interactions – CUT&RUN
## EChO (enhanced chromatin occupancy)

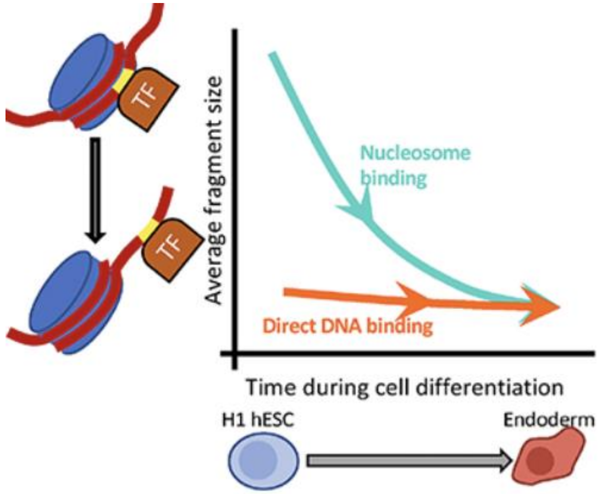
**minimum local fragment size “foci” detection**




**binding overlap analysis**




**direct vs. nucleosomal TF binding**




Meers , Janssens and Henikoff. 2019. Mol Cell

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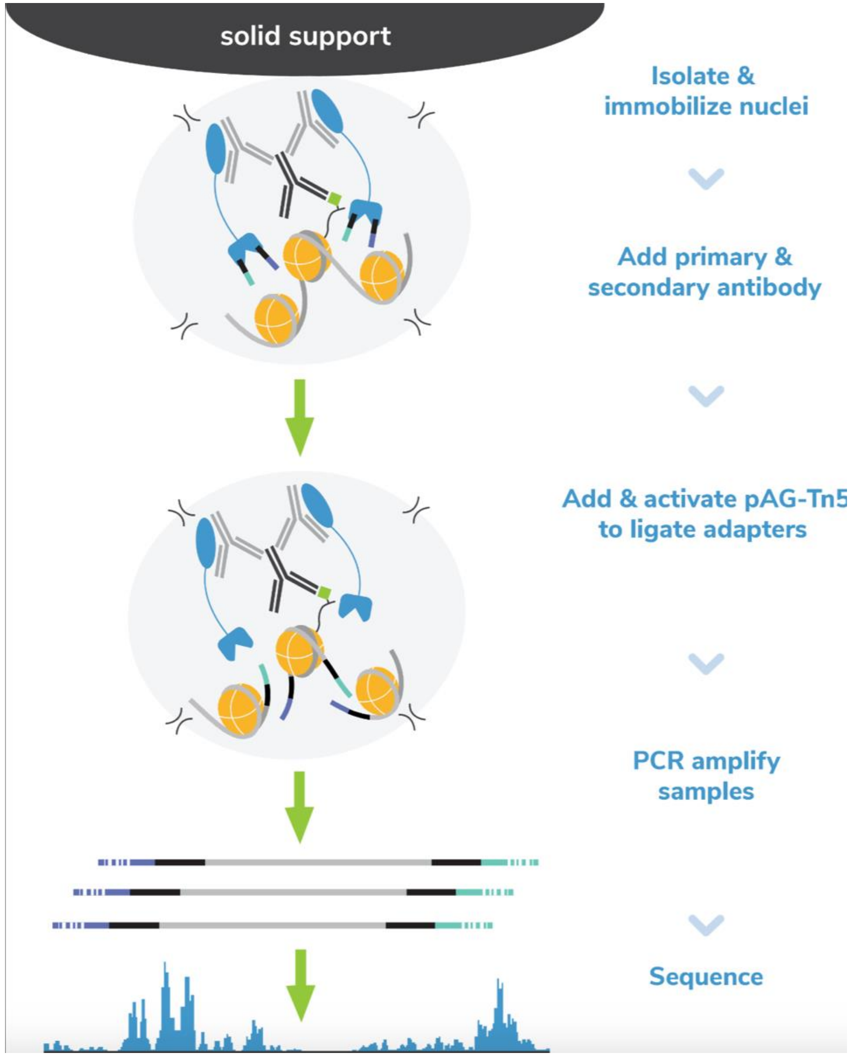


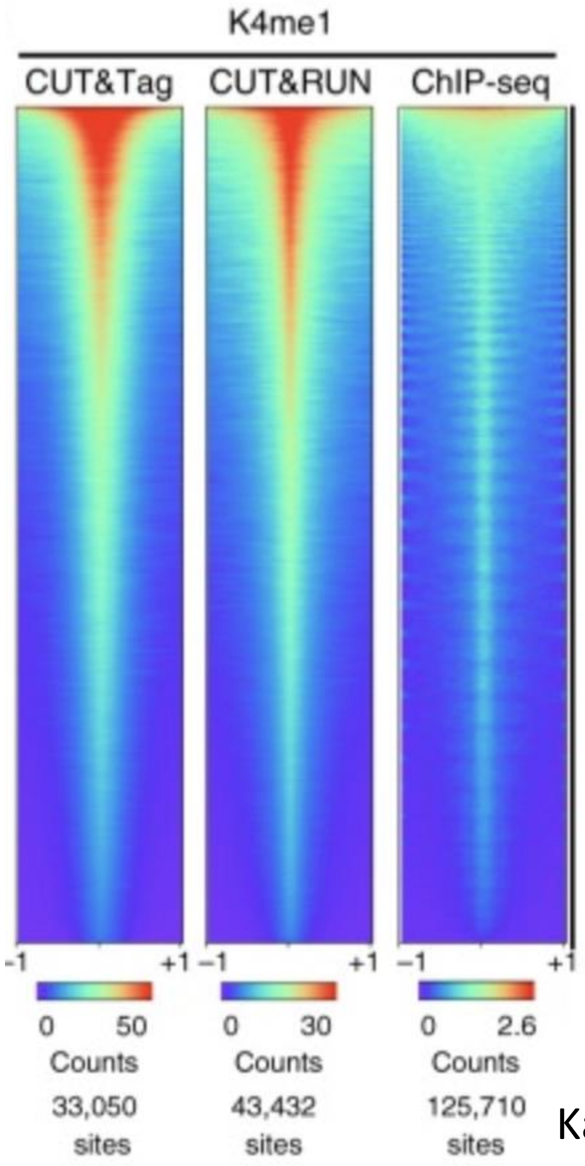


Kaya-Okur, … Henikoff. 2019. Nat Com

Kaya-Okur, … Henikoff. . 2020. Nat Protocols

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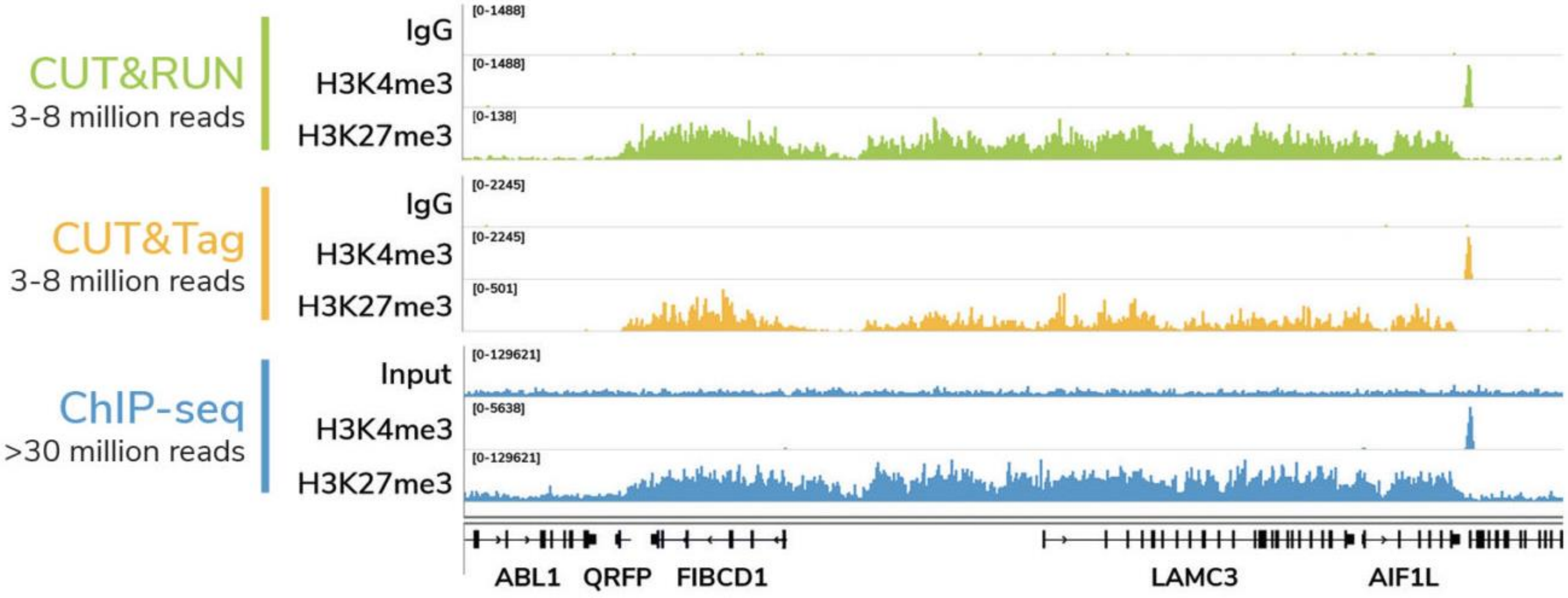

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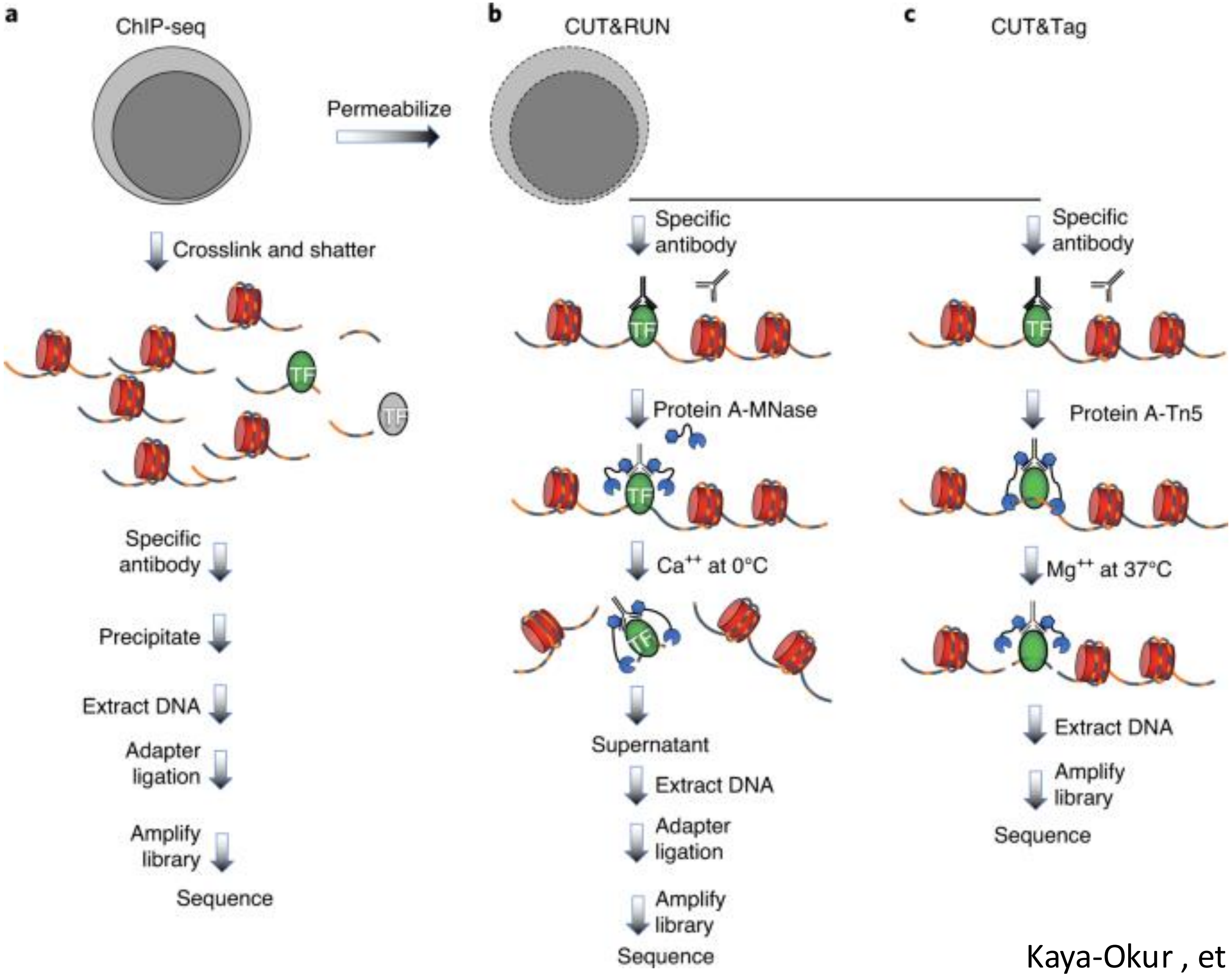


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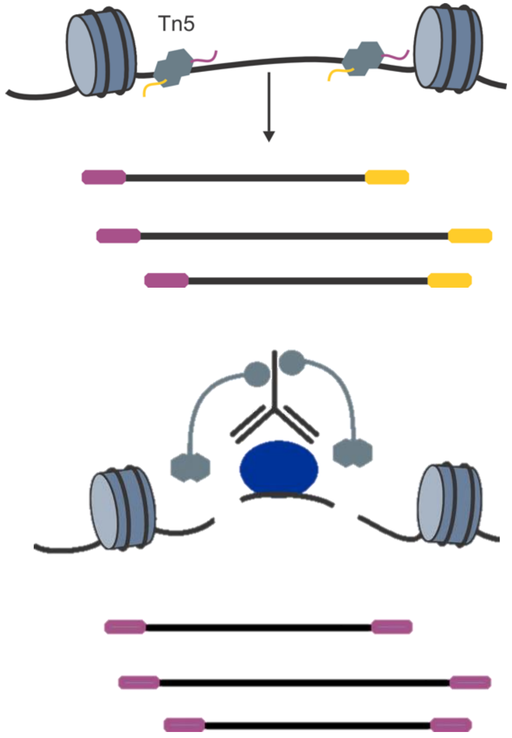


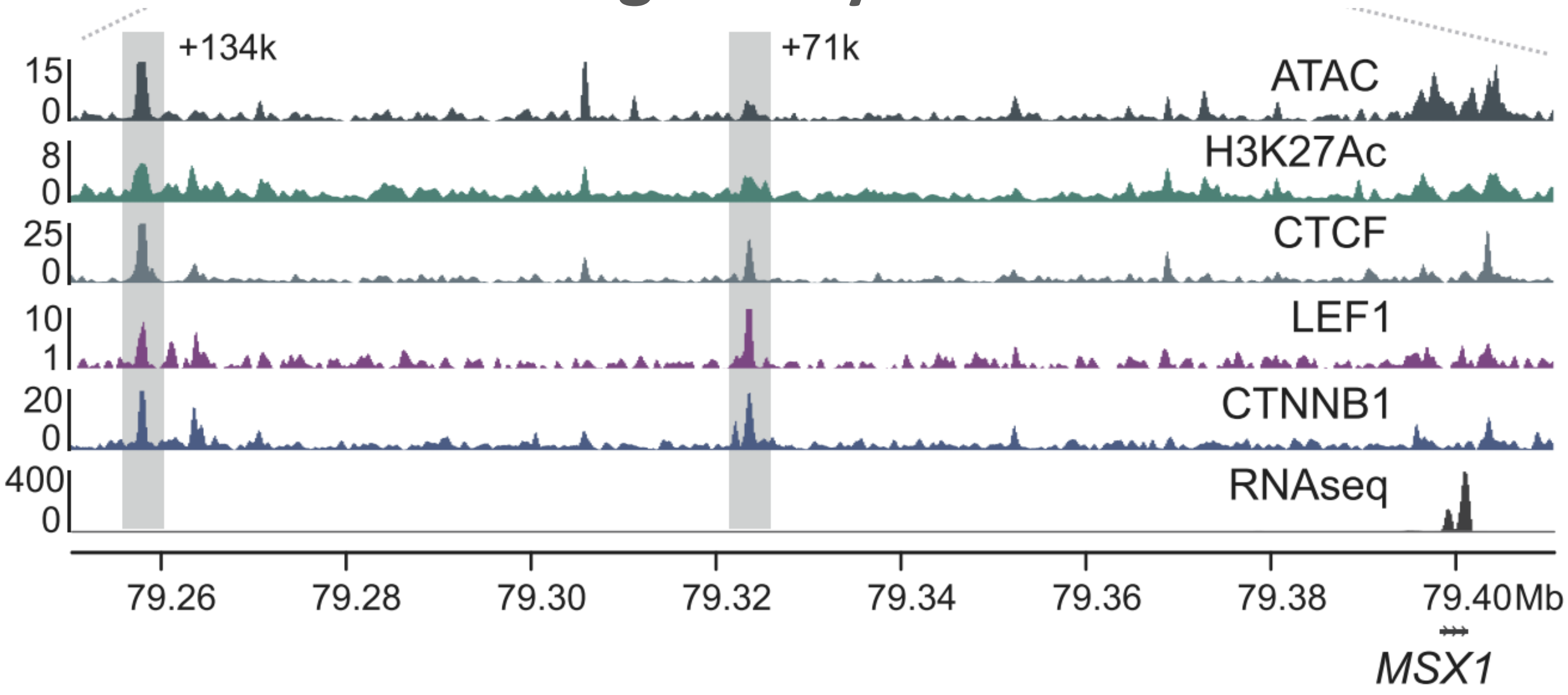


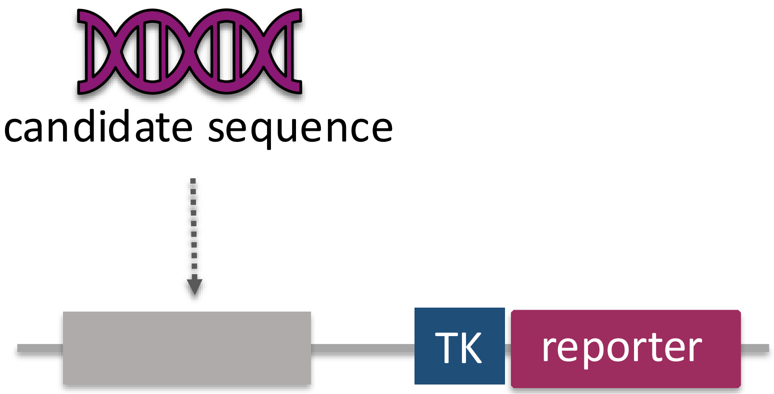


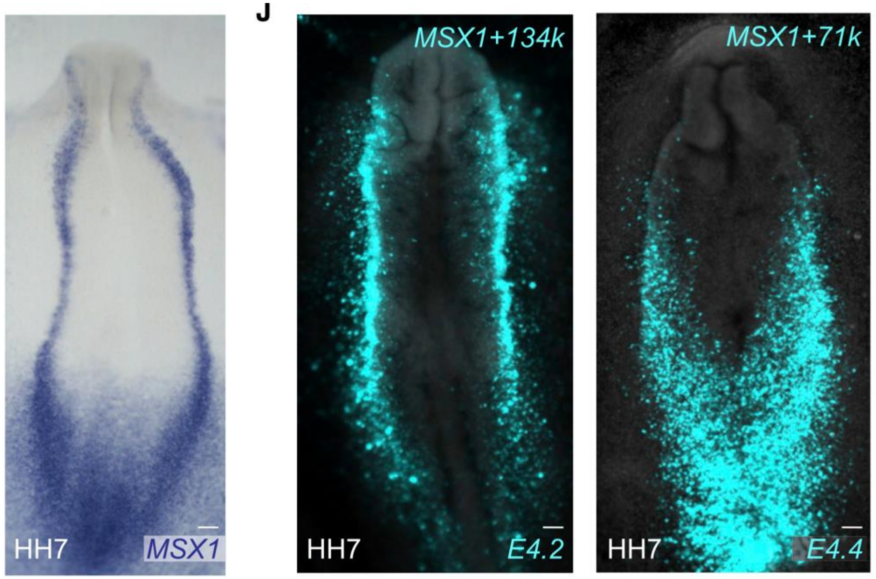


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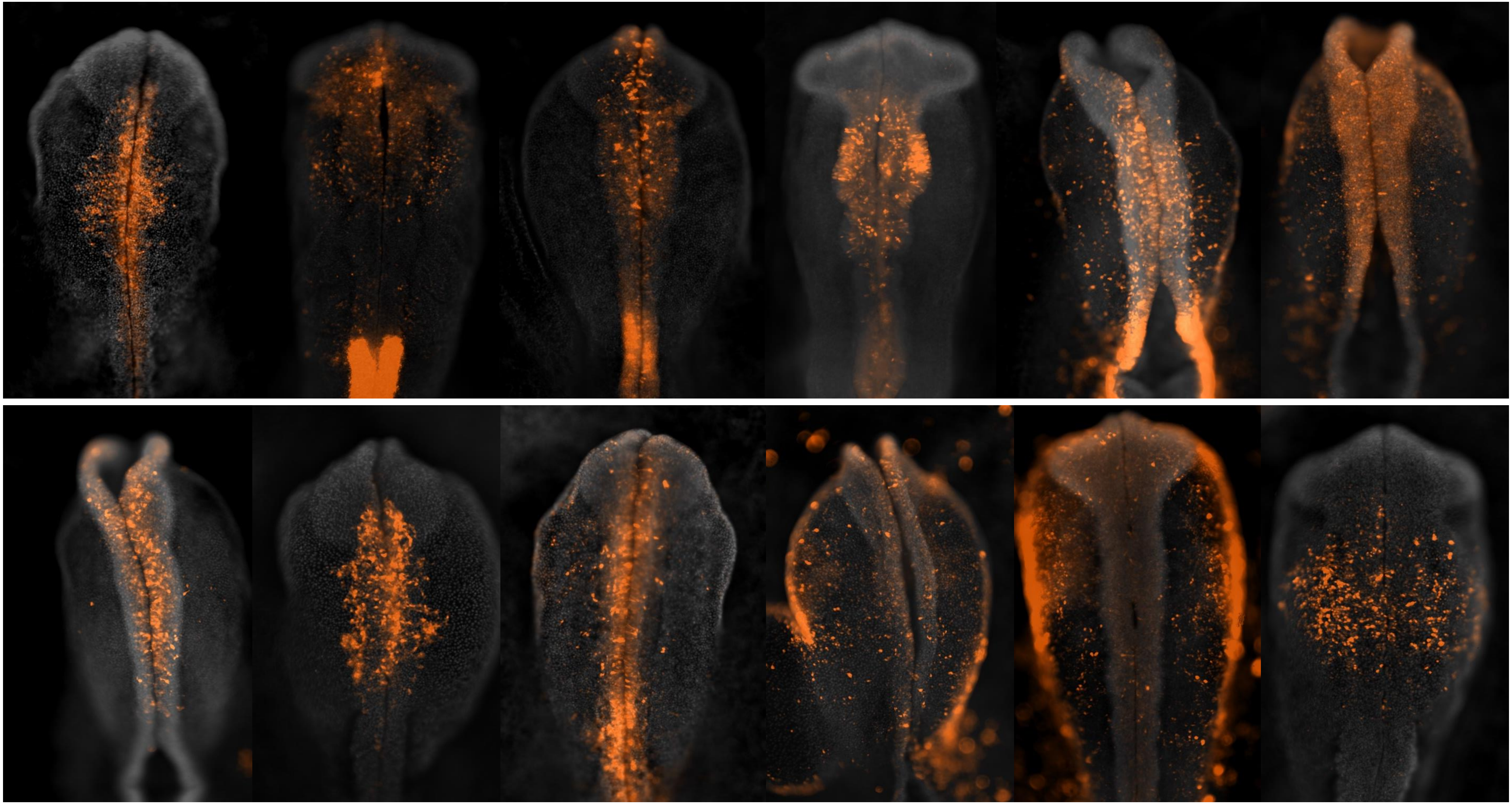
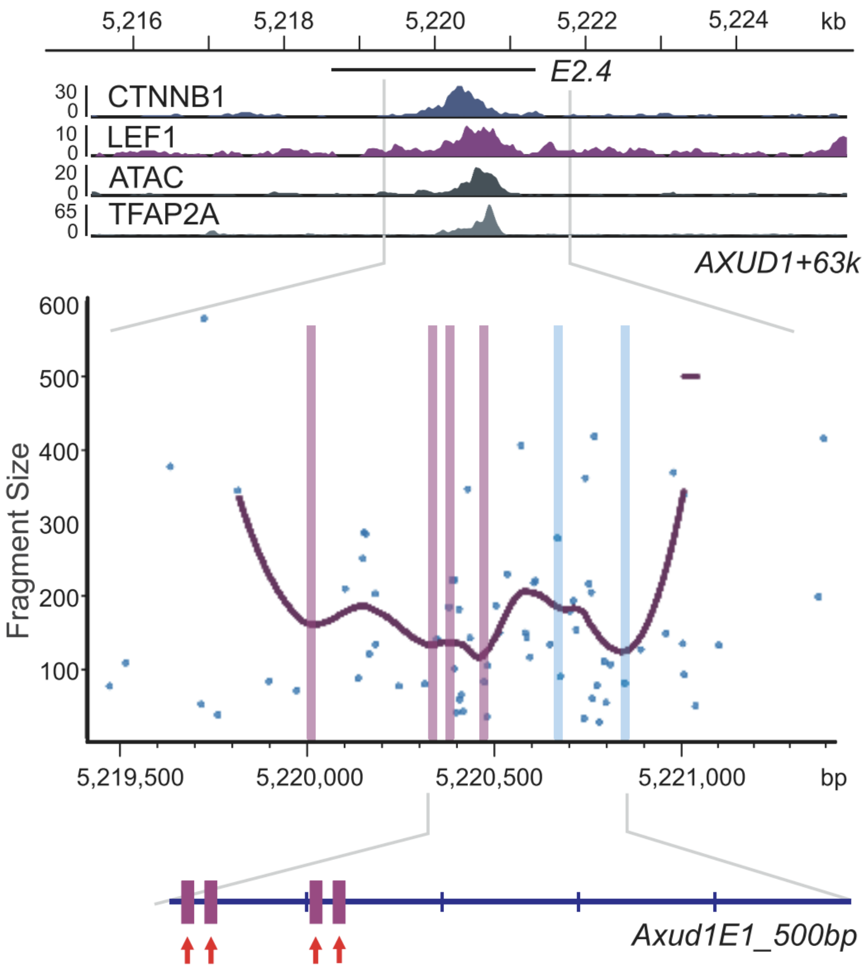
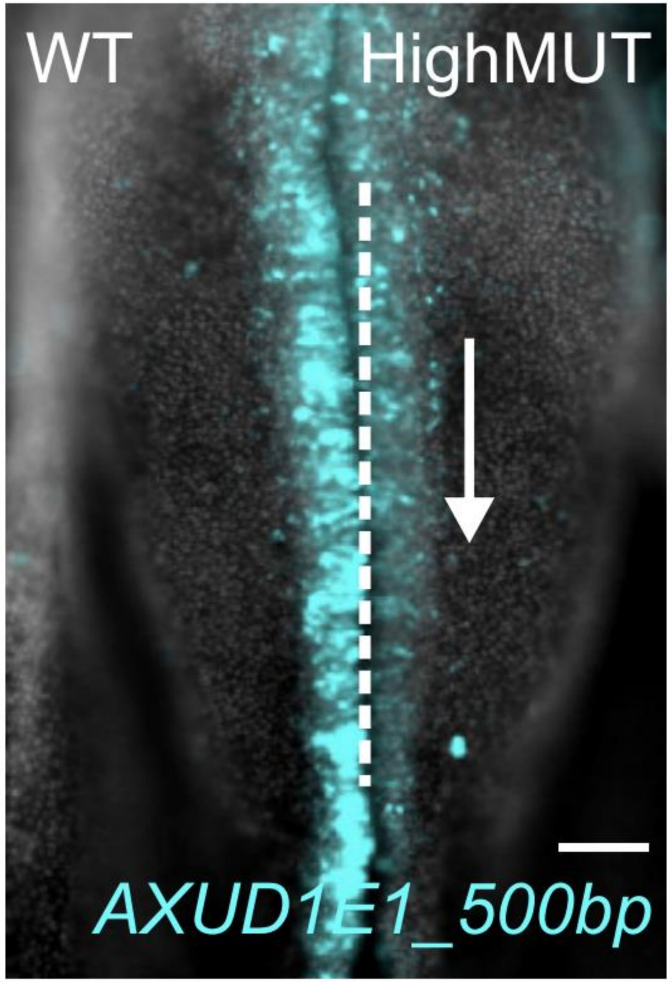


# Cis-regulatory features and how to identify them




database

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